M450 product specification

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1、Product introduction

M450 is a 4G OBD terminal of CATI version (plug-and-play).

4G CAT1 communication module, high precision GPS module, high performance 3-axis gravity sensor and vehicle-mounted ECU computer communication module are adopted here. Vehicle operating status, vehicle condition data, vehicle fuel consumption data, data controlled by drivers and more can be uploaded via wireless module to the cloud platform for vehicle networking management, where vehicle track, driving behaviors, fuel consumption, and information about anti-theft tracking and more can be computed based on the terminal data, thus realizing remote fleet management, and reducing vehicle fuel consumption via the regulation of driving behaviors.

- Support of CAT1 4G network
- Positioning track (Beidou dual mode)
- Vehicle breakdown testing
- Data about driving behavior
- Data about alarms for reminder
- Access to OBD II standard data
- Private protocol data of extendable cars (gauge mileage, remaining oil mass, vehicle VIN, etc.)
- Adaptive passenger vehicles/commercial vehicles/new energy vehicles (voltage range of 9-36V)
- Alarm sounds after it’s pulled out (electric current of 110mA(optional))
- External SIM card
- Support of USB serial port (Micro USB, external serial port for power supply)
- Support of SM2 chip-level data encryption (optional)

2. Product specification

2.1. Appearance structure

Properties	Description
Product appearance	
	
Product size	60*49*24mm(L*W*H)

2.2. Complete machine parameters

Properties	Description
Operating voltage	9V-18V DC
Average operating electric current	180mA@12V
Dormant operating electric current	<10mA@12V
Operating temperature	-30℃～＋70℃
Storage temperature	-40℃～＋80℃
Positioning accuracy	<10m
GPS frequency range	1575.42MHZ
OBD type	Support of OBDII/EOBD
SIM card	External (Micro card)
4G antenna	Internal
GPS antenna	Internal
USB	1. Serial port: 3.3V TTL electrical level 2. Output voltage: 0V output / 5V output / 12V output (You can choose 2 from these 3)

2.3. GPS parameters

Properties	Description
Positioning mode	GPS/BD
Working mode	Single GPSL1, single BD2 B1, GPSL1/BD2 B1 dual mode
Duration for a cold start	≤30s (average)
Duration for a hot start	≤5s (average)
Positioning accuracy	Level<5m | Elevation<10m
Speed measurement accuracy	<0.1m/s
Sensitivity	Capture<-145dBm | Trace<-161dBm
Antenna pattern	External/internal

2.4. CAT1 module

特性

- LTE FDD: Band 1/3/5/8
- LTE TDD: Band 34/38/39/40/41
- GSM: 900/1800MHz
- LCC 31x28x2.4mm
- Operating Temperature: -40°C~+85°C
- Storage Temperature: -40°C~+85°C
- Operating Voltage: 3.3V~4.3V typical 3.8V
- **AT Command Set:** 3GPP TS 27.007 and 27.005, and proprietary FIBOCOM AT commands
- **TX Power**
  GSM900: 32.5 dBm±1dB
  DCS1800: 29.5dBm±1dB
  LTE FDD: 23.0dBm±1dB
  LTE TDD: 23.0dBm±1dB

数据

- LTE FDD: 10Mbps DL/5Mbps UL
- LTE TDD: 10Mbps DL/5Mbps UL
- GPRS: 107Kbps DL/85.6Kbps UL

功能

- MUX: Basic Mode
- SMS: MO/MT Text and PDU modes
- HTTP(S)/FTP(S)/TCP(S)/UDP/IPV4/IPV6/MQTT
- ECM/RNDIS/PPP
- DFOTA/FOAT
- VoLTE/TTS/Recording/FFS/BIP/NITZ/NTP/SSL

2.5. Encryption chip

采用 ARM 32bit CPU 核，提供 7816、SPI 接口，提供硬件 3DES、AES、SM1、SM2、SM3、SM4、PKE 协处理器、硬件 CRC、WDT、真随机数发生器。芯片提供灵活的存储器管理、安全管理机制，支持异常及事件等多种中断机制，提供了多种复位方式和时钟配置方式，并支持低功耗模式。面向移动支付、安全 SE 等安全领域的应用。

CPU 特性

ARM SC000 RISC 处理器

处理器	
CPU	ARM 32bit CPU
系统时钟	36MHz
支持	大端模式，Thumb/Thumb2 指令集
低功耗	支持低功耗模式
产品特性	
FLASH	10 年数据保持时间,10 万次数据擦写; 用户空间 256KB
RAM	40KB
产品特性	支持唯一设备标识
	支持安全密钥存储
	支持对称体系、非对称体系等安全认证
	支持对称加解密、MAC 运算
	支持非对称加解密、HASH 计算等运算
	支持访问数据的线路保护、权限设置等
	支持数据安全存储
	支持安全访问控制
接口	
ISO/IEC7816	兼容 ISO/IEC 7816 T=0 协议
	支持外部时钟 1~5MHz
SPI	符合 SPI 接口规范，最高支持 18MHz
GPIO	6 个

接口特性

符合 ISO/IEC 7816-3 T=0 协议，支持 SPI 接口

安全特性	
存储器数据加密、存储器地址加扰	√
支持存储器访问权限保护机制	√
电压检测/温度检测/频率检测等安全传感器	√
多种防 SPA/DPA/DFA 攻击设计	√
有源屏蔽层	√
多安全算法 DES、3DES、AES、RSA、ECC、SM1、SM2、SM3、SM4、SHA-n	√

电气特性	
工作电压	1.62~5.5V
工作电流	正常电流：小于 10mA
低功耗模式	Standby：小于 200uA @5.5V, 1MHz
ESD	4KV（HBM）
工作温度	-40℃~85℃

应用领域	
移动支付安全 SE	√
TBOX 安全 SE	√
开发环境	
提供 SDK 开发包	√
封装形式	
DFN8(5*6*0.75)	√
QFN32(5*5*0.75)	√

3. Product features

3.1. Functions of vehicle-mounted electric products

Functions of communication management	
Dormancy	The vehicle enters into dormancy after it's misfired for 5 minutes, and the power consumption is reduced.
4G CAT1 network	Support of 4G CAT1 communication
Communication protocol	The data information interacts with the platform with TCP protocol adopted.
Reconnection of network after it's offline	Network that is offline can be automatically reconnected.
Parameter setting	IP address, port number, and APN are set directly in the terminal via the platform, and the password is set for protection.
Positioning function	
Positioning data	It involves latitude, longitude, time, satellite data, speed, direction, ACC status, battery voltage, and OBD speed.
Function of base station positioning	It contains the network base station's LAC and CellID information, and operator code.
Smart track	The terminal can automatically determine the trajectory route to realize intelligent reporting of GPS points, and the butterfly shape effect of trajectory.
Reporting based on time interval	The mode and frequency parameters are set on monitoring platform for sending back the positioning information from the terminal.
Functions of alarm for reminder	
Ignition reminder	When the vehicle is ignited,the ignition alarm information is reported, with alarm time, GPS information involved.
Flameout reminder	When the vehicle is fired off,the flameout alarm information is reported, with alarm time, GPS information involved.
Vehicle alarm for low voltage	When the vehicle voltage is lower than the settable voltage threshold, the low voltage alarm information is reported, with alarm time and GPS information involved.
Alarm for overlong idle speed	When the vehicle voltage is lower than the settable voltage threshold, the low voltage alarm information is reported, with alarm time and GPS information involved.
Alarm for water temperature	For vehicles that support OBD water temperature data, when the water temperature exceeds the settable temperature threshold,
	the water temperature alarm information is reported, with alarm time, GPS information, and water temperature value at that time involved.
Alarm for overspeed	When the vehicle driving speed exceeds the settable speed threshold, the overspeed alarm information is reported, with alarm time and GPS information involved.
Alarm for collision	If a scene in which accelerated speed generated exceeds the settable threshold value when the vehicle is moving, is defined as a serious collision after the driving speeds both before and after are filtered,
	the collision alarm information is reported, with collision time, GPS information, and

	collision accelerated speed involved.
Alarm for the vehicle towing	When the vehicle is towed, the towing alarm information is reported, with towing time, GPS information involved.
Alarm for fatigue driving	When the vehicle driving time is too long, the fatigue driving alarm information is automatically recognized and reported.
Alarm for insertion of device	When the terminal device is inserted, the insertion alarm information is reported.
Alarm for pulling-out of device (optional)	When the terminal device is pulled out, the pulling-out alarm information is reported (when there is a backup electricity option).
Overlong positioning duration	When the positioning time exceeds the settable time threshold in the case of vehicle ignition,
	the alarm information for overlong positioning duration is reported, with alarm time, and GPS information involved.
Driving behaviors	
Driving behaviors	They include start time and terminal time of the driving cycle, total driving cycle mileage, average speed, maximum speed, overspeed duration,
	overspeed times, idle duration, sharp acceleration, sharp deceleration, and sharp cornering.
Driving behavior incidents	Amid vehicle driving process, when incidents of sharp acceleration, sharp deceleration, and sharp cornering occur, these incidents need to be reported separately.
OBD data functions	
Vehicle data flow	The key data flow of the vehicle is reported every 30S, including battery voltage, categories of total mileage, total mileage, total fuel consumption, engine revolving speed, vehicle speed, air mass flow, air inlet temperature, inlet manifold pressure, fault light status, number of fault codes, coolant temperature, vehicle ambient temperature, fuel pressure, atmospheric pressure, valve position sensor, gas pedal position, engine operating duration,fault running mileage, remaining oil mass, engine load, long-term fuel correction (group 1), ignition advance angle, and other data items.
	The exact number of data items varies slightly depending on the vehicle data support.
Reading of vehicle faults	When the terminal identifies the vehicle fault information and the vehicle fault status sees changes, relevant information is reported to the platform.
Clearing of vehicle faults	Vehicle faults are cleared under the instructions issued by the platform.
Remote management functions	
Remote upgrade	Terminal software can be upgraded remotely via GPRS network and FTP server.
Remote restart	The instruction is sent to restart the device via data channel.
Remote inquiry	Via the mobile network on the platform, we can find remotely the terminal information, vehicle type, GPRS communication parameters, heartbeat parameters, GPS/CAN uploading parameters, SIM card information, GPS information, CAN data stream, and current faults. Assorted alarm parameters, and parameters for sharp acceleration, sharp

	deceleration, and sharp cornering.
Remote setup	Via the mobile network on the platform, we can set up remotely the terminal information, vehicle type, GPRS communication parameters, heartbeat parameters, and GPS/CAN uploading parameters, as well as clear vehicle fault, control restart, restore factory settings, and clear blind area data, various alarm parameters, and parameters for sharp acceleration, sharp deceleration and sharp cornering.
Report of the version	Version information of ACC ON is reported each time. Version information of vehicle-mounted electric products is checked under instructions on the platform.
Report of SIM card information	Information about the SIM card IMEI code and more is reported via ACC each time.
	Information of SIM card and IMEI code and more can be checked through the instruction on the platform.
Module self-test	Terminal status self-test: test of whether functional modules (positioning module, Bus module, FLASH, 3D module)
	work normally and whether the fault (not related to wireless communication) alarm of these modules is reported
Backup electricity	
Trace and alarm for backup electricity (optional)	When the terminal is pulled out, and the internal battery is powered, the pulling-up alarm and track location information are reported. (Optional)

3.2. OBD data items

Key OBD data		
Battery voltage	Intake manifold pressure	Gas pedal position
Categories of total mileage	Status of fault light	Engine operating duration
Total mileage	Number of fault codes	Fault running mileage
Total fuel consumption	Coolant temperature	Remaining oil mass
Engine revolving speed	Vehicle ambient temperature	Engine load
Vehicle speed	Fuel pressure	Long-term fuel correction (Group 1)
Air Flow	Atmospheric pressure	Ignition advance angle
Inlet air temperature	Throttle position	Total vehicle running duration

The number of data items supported varies with the information of each vehicle type, and it is subject to the actual support condition of the vehicle.

4. Protocol support

Properties	Description
OBD communication protocol	ISO-15765 500K ST
	ISO-15765 500K EX
	ISO-15765 250K ST

	ISO-15765 250k ex
	ISO-14230 FAST
	ISO-14230 SLOW
	ISO-9141-2
Network communication protocol	TCP protocol

5. Definition of interface

5.1. OBDII connector

OBDII			
Serial number	**Pin definition**	**Serial number**	**Pin definition**
1	NC	9	NC
2	NC	10	NC
3	NC	11	NC
4	GND	12	NC
5	GND	13	NC
6	CAN High	14	CAN Low
7	K Line	15	L Line
8	NC	16	VCC

They are in line with international OBDII standard interface, and wire breaking is not a necessity, with the plug-and-play mode adopted.

5.2. Indicator light

Properties	Description
Indicator light category	① **Green-GPS; ②Blue-OBD; ③Orange-GPRS**
GPRS light	Constant opening of the light indicates that GPRS is online. Quick flashing indicates that the SIM card cannot be found. Slow flashing means that the light is going online. Frequent extinguishing signifies the light is in dormancy or not powered on.
GPS light	Constant opening of it indicates a successful positioning; Slow flashing indicates it’s positioning. Quick flashing explains a module abnormality Frequent extinguishing means it’s in dormancy or not powered on.
OBD light	Blinking indicates the communication with the vehicle is a success. Constant extinguishing indicates there’s no communication with the vehicle or it’s in dormancy or not powered on.

Note: The lights of three colors are circularly switched in accordance with the "orange-green-blue" sequence with 5 seconds as the interval.